

Invigilation Engine: Complexity & Fairness Optimization Audit

A Rigorous Mathematical Analysis and HR Performance Verification Slide Deck

Document Class: Systems Architecture & Computational Analysis

Roster Size: 46 Faculty Members | **Target Architecture:** Local HTTP Integration

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1. Time Complexity: Initial Allocation & Search Refinement

Phase A: Greedy Initial Guess • Mechanism: Chronological sorting of all exam sessions combined with dynamic workload-balanced candidate matching. • Mathematical Bound: $O(S \log S + S * R * F \log F)$ • Implication: Guarantees an initial schedule in fractions of a millisecond.	Phase B: Bounded Hill-Climbing Search • Mechanism: Swaps duty slots and evaluates change using Jain's Index and Gini Coefficient metrics. • Mathematical Bound: $O(I * F)$ (for $I = 20,000$ iterations) • Implication: Runs in absolute linear correlation to roster size. Swift delta updates avoid global re-evaluation.
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2. Backtracking, Unified Scaling, & Glossary

Phase C: Bounded Backtracking

- **Mechanism:** Deep tree search to resolve complex room allocation conflicts.
- **Mathematical Bound:** Exponential $O(F^K)$ worst-case; capped at 100,000 steps.

Phase D: Combined Unified Model

- **Pipeline Complexity:** $O((S \log S + S * R * F \log F) + I * F + 100,000 * F)$
- **Actual Scaling:** Linear $O(S)$ on production data because SF and IS are fixed constants.

Mathematical Notation Glossary

Symbol	Definition
S	Total exam sessions
F	Roster size (46 members)
R	Invigilators per room
I	Search steps (fixed at 20,000)
K	Unassigned remaining slots

3. Real-Time System Local Runtime Performance

Backend Execution Time Ranges (Locally)

- **Greedy Phase:** Runs in less than **5 ms** to construct the initial allocation guess.
- **Local Search Optimization Phase:** Takes between **50 ms to 150 ms** to complete the full 20,000 improvement swaps.
- **Worst-Case Backtracking Governor:** Takes between **1.0 to 2.0 seconds** to run the maximum 100,000 steps on highly constrained/unfeasible rosters.
- **Total Local Execution Range:** **50 ms to 2,000 ms (2.0 seconds max)**, ensuring backend responsiveness and avoiding page or thread hangs.